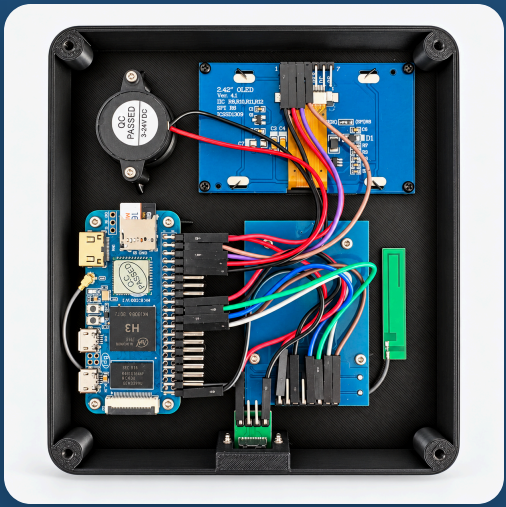
ICUBE DEVELOPMENT (CALGARY) LTD.

Timeclock Build and Install Guide

Step by step, from printing the case to a working clock on the wall. No previous experience needed. Follow every step in order.





START HERE

Read this first

What you'll do

PART	WHAT HAPPENS	CHAPTER
A. Build	Print the case, solder the parts, put them in the case and connect the wires	1 to 4
B. Software	Prepare the memory card, start the clock, copy the timeclock software to it and install it	5 to 9
C. Finish	Create the first manager, test, close the case and hang it on the wall	10 and 11

How to read this guide

- **Bold words** are buttons, labels or messages you'll see.
- Text in a grey box like the one below is something you **type exactly**, including spaces and capital letters. Press **Enter** at the end of each line.

  ```
  cd /root
  ```

- Do every step in order. Don't skip ahead.
- Tick the boxes ☐ as you go.

Safety

> **Never connect power until the guide tells you to**
> Wiring with power connected can destroy the board. Each time you change a wire, unplug the USB cable first.

- A soldering iron is about 350 °C. Use a stand, work on a heat-proof surface and keep the room ventilated.
- Touch a metal object before handling the boards, to discharge static. Hold boards by their edges.

You'll need some help with

- The **Wi-Fi name and password** the clock will use. It must be a 2.4 GHz network.
- Access to your **router's admin page** (to find the clock's address), or someone from IT who has it.
- For the first clock, the **manager's name, work email** and a new card for them.

START HERE, CONTINUED

Shopping list

☐	PART	QTY	NOTES
☐	Banana Pi BPI-M2 Zero	1	Comes without pins (bare holes)
☐	microSD card	1	16 GB or larger, good brand
☐	2.42" SSD1309 OLED display, 128 × 64	1	The 7-pin module shown on the Components page
☐	RC522 RFID reader kit	1	Includes a test card and key fob
☐	Active buzzer with mounting ears, 22.6 mm	1	2 wires (red and black). Must work from a 3.3 V signal
☐	USB-C breakout board (pads G, D+, D−, V)	1	The type shown on the Components page
☐	Wi-Fi antenna, IPEX / u.FL plug	1	Flat stick-on type
☐	Right-angle (90°) male pin header, 2.54 mm	1 strip	Enough for 40 (2 rows of 20) + 8 + 7 + 2 pins
☐	Female-to-female jumper wires (Dupont), 10 to 20 cm	14	Different colours make checking easier
☐	M2 × 5 mm self-tapping screws	8	OLED (4) and RFID reader (4)
☐	M2 × 6 mm self-tapping screws	6	Banana Pi (4) and USB-C board (2)
☐	M2 × 8 mm self-tapping screws	2	Buzzer
☐	M3 × 12 mm countersunk (flat head) self-tapping screws	4	Rear cover
☐	USB charger (5 V, 2 A or more) and USB-A to USB-C cable	1	A normal phone charger with the big USB-A socket is ideal
☐	2 wall screws and anchors	2	Pan head. Head no wider than 8 mm, shank no thicker than 4.3 mm
☐	Heat-shrink tube, small	a few cm	For the buzzer wire joins

Tools

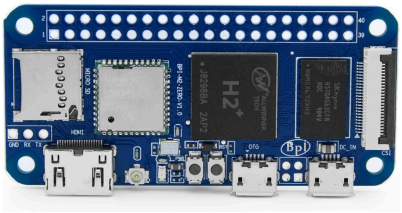
3D printer · soldering iron, solder and flux · side cutters · small Phillips screwdriver · tweezers · a computer with a microSD card reader · a spirit level, drill and pencil for wall mounting.

Screw lengths come from the case design. If a screw feels like it's bottoming out, stop. Don't force it.

START HERE, CONTINUED

Components

Check you have each of these and that they look like the photos.



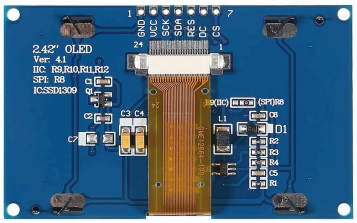
Banana Pi BPI-M2 Zero

The small computer that runs the clock. The 40 holes along one edge are the "header". You'll solder pins into them.



RC522 card reader

Reads the staff cards. Its 8 holes are labelled SDA, SCK, MOSI, MISO, IRQ, GND, RST, 3.3V. Keep the test card and fob.



OLED display, back

7 holes labelled 1 GND, 2 VCC, 3 SCK, 4 SDA, 5 RES, 6 DC, 7 CS. It needs a small change before use (Chapter 2).



OLED display, front

The glass shows through the case window. Don't press on the glass or bend the orange ribbon.



Buzzer

Beeps on every tap. Red wire is + (plus), black is − (minus). The two ears take screws.



USB-C board

Where the power cable plugs in. Only the **V** and **G** holes are used. D+ and D− stay empty.

CHAPTER 1

Print the case

Unzip `timeclock-case-v8.34-STL.zip`. You need these files:

☐	FILE	WHAT IT IS
☐	`timeclock_v8_34_oled_fit_test.stl`	A small test piece. **Print this first.**
☐	`timeclock_v8_34_front_PRINT.stl`	The front of the case
☐	`timeclock_v8_34_back.stl`	The rear cover
☐	`timeclock_v8_34_buzzer_clamp.stl`	Only if your buzzer has no ears

Slicer settings

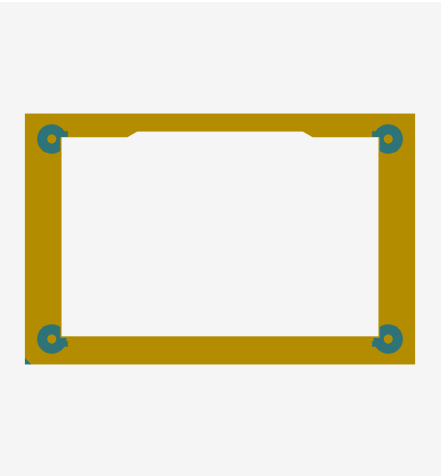
1. Open your printer's slicer (for example Bambu Studio, PrusaSlicer or Cura) and load the file.
2. **Don't rotate or flip the parts.** They're already the right way up: the front lies face-down, and the cover lies with its flat outside face down.
3. Set: layer height **0.24 mm**, nozzle **0.4 mm**, **supports off**, 3 walls, 20 % infill. Use black PETG or PLA.
4. Slice and print.

Check the test piece

From the back, press the OLED's glass into the window of the test piece.

- ☐ The glass goes in without forcing and sits level with the front face.
- ☐ The 4 corner holes of the OLED line up with the 4 small holes in the test piece.

If it's too tight or the holes don't line up, don't print the front yet. Ask iCube for help with your printer's accuracy.



OLED test piece

Then print the front and cover

After printing, pick off any strings of plastic inside the window and buzzer holes. Check the cover fits snugly over the front's rim all the way round.

CHAPTER 2

Solder the parts

The pins must be **right-angle (bent 90°) pins**. Straight pins stick up too far and the case won't close.

1. Banana Pi: 40 pins

1. Cut a 2 × 20 piece of right-angle header (or two 1 × 20 strips).
2. Push the short ends into the 40 holes from the **top** of the board (the side with the big black chips). The long ends must point **away from the board**, out past its edge.
3. Turn the board over and solder every pin on the underside. Each joint should be a small shiny cone. Make sure no blob of solder touches a neighbouring pin.

2. RC522 reader: 8 pins

Use right-angle pins, not the straight ones in the kit. Push the short ends into the 8 holes from the **plain back** of the reader (the side without the printed antenna coil), long ends pointing **away from the board**. Solder on the antenna side.

3. OLED display: 7 pins

Push the short ends in from the **back** (the side with the printing), long ends pointing **inward, over the back of the board**, like the reference photo. There's no room above the top edge of the display inside the case. Solder from the front edge, keeping the iron away from the glass.

4. USB-C board: 2 pins

Solder right-angle pins into **V** and **G** only, so they point along the board, away from the socket.

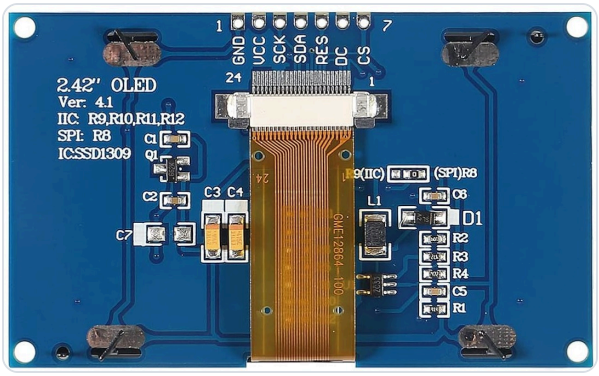
5. Buzzer wires

Cut one end off two jumper wires. Strip about 5 mm from the buzzer wires and the jumper wires, twist red to one and black to the other, solder, and cover each join with heat-shrink. Label or remember which is red (+).

Change the display to I2C mode

The display comes set for a different connection type (SPI). The clock needs it set to **I2C** (printed **IIC** on the board). You change it with five tiny positions on the back, each marked with an R number.

☐	POSITION	WHAT TO DO
☐	R8	Remove the tiny 0 Ω resistor (a small part marked "0"), leaving the pads open
☐	R9	Solder that removed resistor here, or join the two pads with a blob of solder
☐	R10	Join the two pads with solder
☐	R11	Join the two pads with solder
☐	R12	Join the two pads with solder



The printing "IIC: R9, R10, R11, R12 / SPI: R8" is at the top left. The R9/R8 position is to the right of the orange ribbon.

How

- To remove R8: add a little fresh solder, then heat both ends at once with the iron and slide the part off with tweezers.
- To join pads: put a little flux on, touch the iron with a small amount of solder across both pads, then lift away. The two pads should be joined by one shiny bridge.
- R9 sits next to R8 by the connector. On the board it's printed **R9(IIC) ▭ (SPI)R8**.

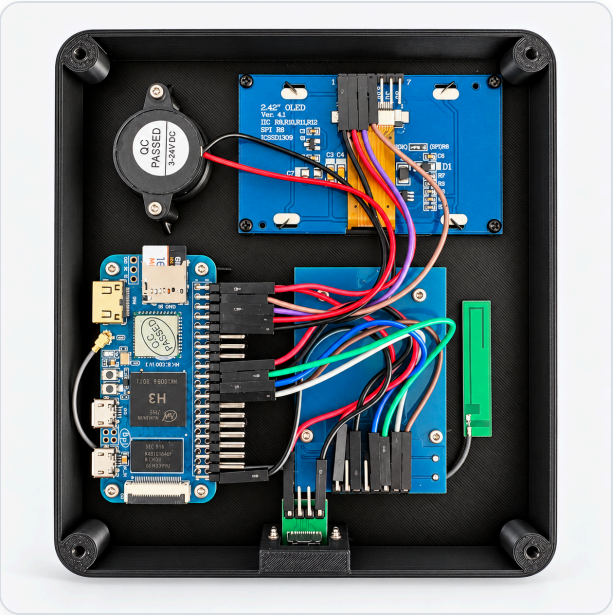
> **Small parts**
> These are very small. Use good light and tweezers. If you're not comfortable, ask someone experienced to do this step.

CHAPTER 3

Put the parts in the case

Lay the front face-down on a soft cloth with the open back facing you. Match each part to the reference photo. Turn the screws in gently and stop as soon as they're snug, because they cut their own thread in the plastic.

1. ☐ **USB-C board** (bottom edge, middle). From **outside** the case, slide it into the slot so the socket faces down and the board sits flat with the case. Fix it with 2 × M2 × 6 mm.
2. ☐ **Display** (top right in the photo). Press the glass into the window from inside. Fix it with 4 × M2 × 5 mm into the 4 corner holes. Its pins point down over the board.
3. ☐ **Card reader** (below the display). Antenna side facing the front of the case, pins at the bottom. Fix it with 4 × M2 × 5 mm.
4. ☐ **Buzzer** (top left). Drop it into the round ring. Fix it with 2 × M2 × 8 mm through its ears.
5. ☐ **Banana Pi** (bottom left). Chips facing you, microSD slot at the top, pins on the side facing the middle of the case. Fix it with 4 × M2 × 6 mm.
6. ☐ **Wi-Fi antenna.** Press its tiny plug onto the antenna socket on the Banana Pi until it clicks (Chapter 4 shows where). Peel and stick the antenna to the inside wall on the right.

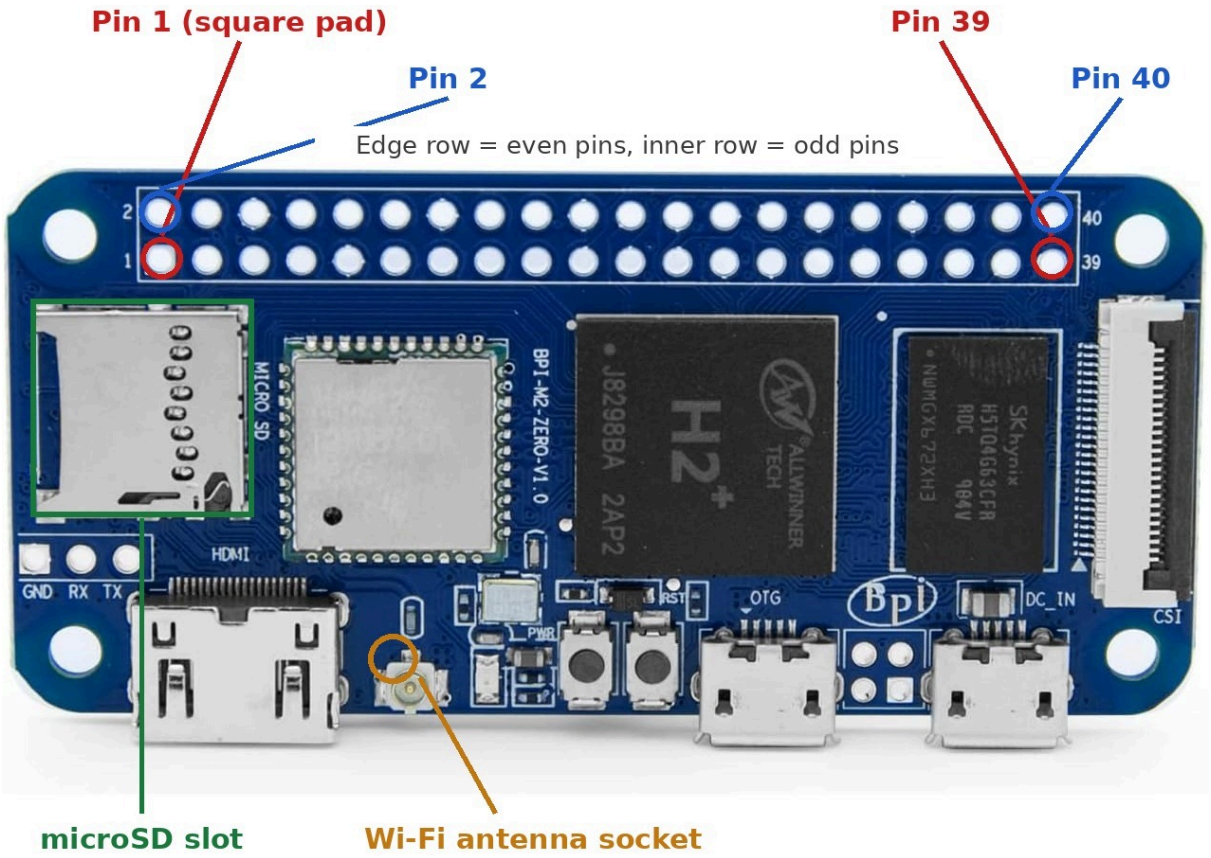


What it should look like (from the open back)

> **Leave the rear cover off**
>
> You'll put the memory card in later, and you need to see inside for testing.

CHAPTER 4

Find the pins on the Banana Pi



The board has 1, 2, 39 and 40 printed next to the header

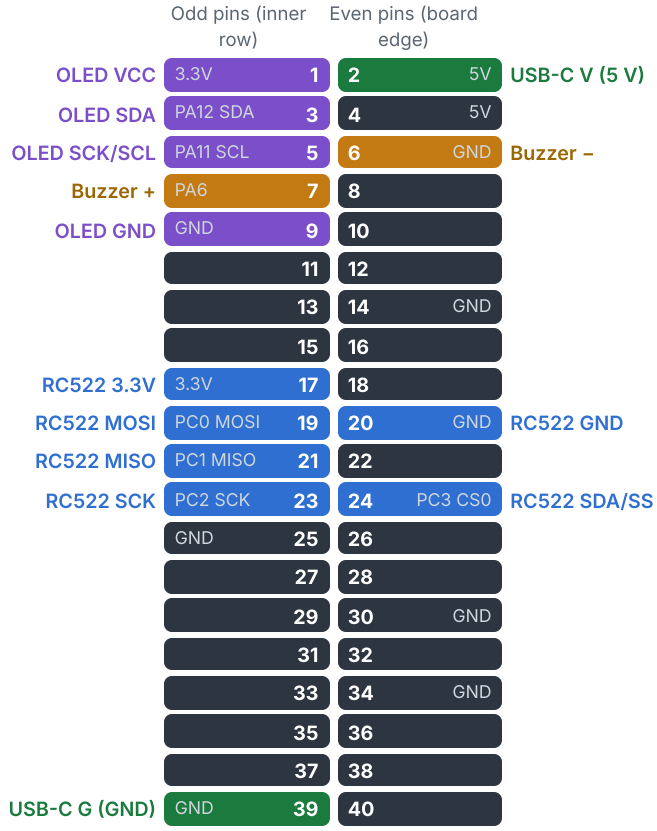
- Pin **1** has a **square** pad and is at the end nearest the microSD slot.
- The row along the **edge of the board** has the **even** pins: 2, 4, 6 … 40.
- The **inner** row has the **odd** pins: 1, 3, 5 … 39.
- To find a pin, count along its row in twos from the nearest printed number. For example, pin 24 is the 12th pin along the edge row from pin 2.

> **Tip**
> Before wiring, put a small dot of marker next to pins 1, 9, 17, 24 and 39 as landmarks.

Connect the wires

The USB cable must be **unplugged**. Push each jumper firmly onto its pin. Do them in this order and tick each one.

☐	#	FROM (LABEL ON THE PART)	TO BANANA PI PIN
Card reader (RC522)			
☐	1	SDA	**24**
☐	2	SCK	**23**
☐	3	MOSI	**19**
☐	4	MISO	**21**
☐	5	GND	**20**
☐	6	3.3V	**17** (never 2 or 4)
		IRQ and RST	Leave empty
Display (OLED)			
☐	7	1 GND	**9**
☐	8	2 VCC	**1**
☐	9	3 SCK	**5**
☐	10	4 SDA	**3**
		5 RES, 6 DC, 7 CS	Leave empty
Buzzer			
☐	11	Red (+)	**7**
☐	12	Black (–)	**6**
USB-C board			
☐	13	V	**2**
☐	14	G	**39**



Every other pin stays empty.

Check the wiring

Mistakes here can destroy parts, so check slowly. Ask a second person to check too.

☐	CHECK
☐	You count exactly **14** wires on the Banana Pi pins.
☐	Pin **2** goes only to the USB-C board's **V**. Nothing else is on pins 2 or 4 (they carry 5 V).
☐	The card reader's **3.3V** wire is on pin **17**, and the display's **VCC** wire is on pin **1**.
☐	The buzzer's **red** wire is on pin **7** and **black** on pin **6**.
☐	Each plug sits fully on one pin, not between two pins or off the end of a row.
☐	No bare metal or solder touches anything it shouldn't.
☐	The wires lie flat and don't press on the display ribbon.

> **Still don't plug in power**
> Power goes on in Chapter 6, after the memory card is ready.

CHAPTER 5

Prepare the memory card

Do this on your computer.

1. Download the clock's system

1. Go to **github.com/mkinderwater/bpi-zero-wbuild/releases/tag/3.13**.
2. Scroll down to **Assets**. Click **bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz** (about 148 MB) to download it.
3. Don't open or unzip it.

2. Write it to the card

1. Install **balenaEtcher** from balena.io/etcher and open it.
2. Put the microSD card in your computer's card reader.
3. Click **Flash from file** and choose the **.img.gz** file you downloaded.
4. Click **Select target** and tick your microSD card. Check the size matches your card, so you don't erase the wrong drive.
5. Click **Flash!** and wait until it says **Flash Complete**.

> **Windows: if a message says "You need to format the disk", click Cancel**
> Formatting erases the card and you'd have to start again. Windows may show this more than once. Always click Cancel.

3. Open the settings file

1. Take the card out and put it back in.
2. A drive called **BPIWBUILD** appears. Open it.
3. Open the file **CONFIG.TXT**:
   - Windows: right-click it and choose **Open with > Notepad**.
   - Mac: open it with **TextEdit**. If you see formatting buttons, choose **Format > Make Plain Text**.

Fill in the settings

Scroll to the bottom of CONFIG.TXT. Lines starting with **#** are only notes, so leave them alone. Change only the six lines at the bottom so they look like this, with your own details:

```
SSID=OfficeWiFi
PSK=wifi-password-here
COUNTRY=CA
HIDDEN=false
TIMEZONE=America/Edmonton
ROOT_PASSWORD=Choose-A-Strong-Password-26
```

LINE	WHAT TO PUT AFTER THE =
SSID	The Wi-Fi network name, exactly as it appears on a phone (capitals matter)
PSK	The Wi-Fi password (8 to 63 characters)
COUNTRY	CA for Canada, US for the United States
HIDDEN	false, unless IT tells you the network is hidden (then true)
TIMEZONE	Your area, copied from the list in the file, for example America/Edmonton, America/Vancouver or America/Toronto
ROOT_PASSWORD	A new password for the clock, 8 to 64 characters. **Required.** Without it you can't get into the clock

- **No spaces** before or after the **=**, and **no quote marks**.
- **Write down the ROOT_PASSWORD now** and keep it safe. The clock deletes it from this file after it starts, and you'll need it in Chapter 7.

1. Save the file (Ctrl+S on Windows, Cmd+S on Mac).
2. Eject the card properly: Windows, right-click the drive and choose **Eject**. Mac, drag it to the Trash.

CHAPTER 6

Start the clock

1. ☐ Slide the microSD card into the Banana Pi's slot, **gold contacts facing the board**, until it clicks.
2. ☐ Plug the USB-A to USB-C cable into the charger, then into the clock's USB-C socket on the bottom of the case.
3. ☐ A small **red light** on the Banana Pi comes on. If nothing lights up, unplug and recheck wires 13 and 14.
4. ☐ **Wait 10 minutes.** On its first start the clock sets itself up and joins the Wi-Fi. The display stays blank during this time. That's normal.

Find the clock's address

Your computer needs the clock's network address (its **IP address**, four numbers like **192.168.1.50**).

1. Open your router's admin page (or ask IT) and look at the list of connected devices.
2. Find the new device that appeared in the last few minutes. Its name is a mix of letters and numbers.
3. Write down its IP address: _____ . _____ . _____ . _____
4. In the router, choose **Reserve** (also called **static** or **fixed IP**) for this device, so its address never changes.

A phone app such as Fing can also list devices on your Wi-Fi. If you connect a monitor to the Banana Pi's mini-HDMI port, the screen shows **Wi-Fi: wlan0 MAC: … IPv4: …** with the address.

> **Not on the list after 15 minutes?**
> Unplug the power. Put the card back in your computer and check CONFIG.TXT: the Wi-Fi name and password exactly right, a 2.4 GHz network, and ROOT_PASSWORD filled in. Fix it, save, eject and try again.

CHAPTER 7

Connect to the clock

You'll type commands to the clock from your computer. This is called **SSH**. It's built into Windows 10, Windows 11 and Mac.

1. Open a command window:
   - Windows: click Start, type **PowerShell** and press Enter.
   - Mac: open **Applications > Utilities > Terminal**.
2. Type this, using your clock's IP address instead of 192.168.1.50, and press Enter:

   ```
   ssh root@192.168.1.50
   ```
3. The first time, it asks **Are you sure you want to continue connecting (yes/no)?** Type **yes** and press Enter.
4. It asks for a **password**. Type the ROOT_PASSWORD from Chapter 5 and press Enter. **Nothing appears on screen while you type the password.** That's normal. Just type it and press Enter.
5. You're in when the line ends with **#**, for example **root@...:~#**.

Check it's the right system. Type:

```
cat /etc/bpi-zero-wbuild-release
```

The answer should include **3.13-trixie-minimal-hw**.

> **Copy and paste**
> You can copy commands from this guide. To paste into PowerShell, right-click. On Mac, press Cmd+V. Check the pasted text before pressing Enter.

To leave the clock's command window later, type **exit**. Keep it open for now.

CHAPTER 8

Copy the software to the clock

You need the file **mk-timeclock-lite-RC3-cleanup6.zip** on your computer. Pick **one** of these ways to copy it.

Windows: WinSCP (easiest)

1. Install **WinSCP** from winscp.net and open it.
2. In the login box: File protocol **SFTP**, Host name = the clock's IP address, Port **22**, User name **root**, Password = your ROOT_PASSWORD.
3. Click **Login**. If asked about a host key, click **Accept**.
4. The right side shows the clock. Make sure it's in the folder **/root**.
5. Drag the zip file from the left side (your computer) to the right side.

Mac, or Windows PowerShell

1. Open a **new** Terminal or PowerShell window (keep the SSH one open).
2. Go to the folder with the zip file. If it's in Downloads, type:

```
cd Downloads
```

3. Then type (using your clock's IP address):

```
scp mk-timeclock-lite-RC3-cleanup6.zip
root@192.168.1.50:/root/
```

4. Type the ROOT_PASSWORD (invisible again) and press Enter. Wait for **100%**.

CHAPTER 9

Install the timeclock software

Go back to the SSH window from Chapter 7 (the one ending in **#**). Type each line, press Enter, and wait for the **#** to come back before typing the next.

☐	TYPE THIS	WHAT IT DOES
☐	`apt-get update`	Gets the latest list of software (about a minute)
☐	`apt-get install -y unzip python3`	Installs the unzip tool
☐	`cd /root`	Goes to the folder you copied the zip into
☐	`unzip mk-timeclock-lite-RC3-cleanup6.zip`	Unpacks the software
☐	`cd mk-timeclock-lite-RC3-cleanup6`	Goes into the unpacked folder
☐	`python3 rc1-preflight.py`	Checks the files. Must show **PASS**

Now install. Choose one

Your first (or only) clock: Primary

```
./install-master.sh
```

It takes **about 7 minutes**, longer on slow internet. Lots of text scrolls past. Don't close the window or unplug the clock. It's finished when the last line says **Installed role: primary**.

Any extra clock: Secondary

```
./install-slave.sh
```

It ends with **Slave installed. Adopt at http://<clock-ip>:8090/**. Then a manager joins it to the Primary (see "Adding a Secondary clock" in Chapter 10).

Use only one of these on each clock, once

Never run both on the same clock. If you chose the wrong one, stop and contact iCube.

If anything ends with an error instead, copy the last 20 lines of text and send them to iCube.

CHAPTER 10

Create the first manager

Primary clock only. Use a browser on a computer connected to the same network.

1. Go to **https://192.168.1.50/setup** (your clock's IP address).
2. The browser shows a security warning. That's expected at this stage, because the clock uses a temporary certificate.
   - Chrome or Edge: click **Advanced**, then **Proceed to ...** or **Continue to ...**
   - Safari: click **Show Details**, then **visit this website**.
   - Firefox: click **Advanced**, then **Accept the Risk and Continue**.
3. Fill in the manager's name, work email, position and a password of 12 or more characters. Submit the form.
4. The page says **Tap the new manager card on the clock within 30 seconds.** Tap a **new, unused card** on the clock's tap area. That card is now the manager card.
5. Sign in at **https://192.168.1.50/login**.

> **Manager card and attendance card are different**
> Give the manager a second card for their own attendance. The Manager Guide explains everything from here.

Adding a Secondary clock

1. On the Primary's website, go to **Device > Clock management** and click **Generate code**.
2. In a new browser tab, open **http://<secondary-clock-ip>:8090/** and enter the code within 10 minutes.
3. Back on Clock management, approve the new clock under **Waiting for approval**.

For IT, later

So email links work without warnings, install a trusted certificate under **Settings > HTTPS certificate**. Set up **Settings > Outgoing email** so the clock can send email.

CHAPTER 11

Test, close and hang it

Test



☐

Display works
Shows the date, time and TAP YOUR CARD.



☐

Manager card
Tap it: HELLO MANAGER, and a beep.



☐

Test card from the kit
Tap it: INVALID CARD, NOT RECORDED. That's correct, because it isn't registered.

If the clock says **CLOCK NOT SYNCED**, it can't get the correct time from the internet yet. Wait 5 minutes and try again, then ask IT.

Close the case

1. Unplug the power.
2. Check no wire is pinched at the edge. Fit the rear cover over the rim, with the vent slots behind the Banana Pi and the two keyholes at the top.
3. Fix it with 4 × M3 × 12 mm flat-head screws. They sit flush with the back.

Hang it on the wall

1. Print the **drilling template** on the last page at **100 % / Actual size**, not "fit to page". Check the 80 mm line measures 80 mm with a ruler.
2. Tape it to the wall at a comfortable height, level, near a power socket. Keep away from large metal surfaces and direct sunlight.
3. Drill at the two crosses, fit the anchors and screw in the two screws, leaving the heads about **3 mm** out from the wall.
4. Hold the clock so the screw heads go through the round part of the keyholes, then slide it **down** until it stops.
5. Plug in the USB cable from below and wait for TAP YOUR CARD.

HELP

If something goes wrong

WHAT YOU SEE	WHAT TO DO
No red light on the Banana Pi	Unplug. Check wires 13 (V to pin 2) and 14 (G to pin 39). Try another charger and cable.
The clock never appears on the router list	Check CONFIG.TXT (Chapter 5): Wi-Fi name and password exactly right, 2.4 GHz network, ROOT_PASSWORD filled in. Wait 10 minutes after power on.
Permission denied when typing the password	The user name must be **root**. Retype the ROOT_PASSWORD carefully (it's invisible as you type). Capitals matter.
ssh is not recognized (Windows)	Use WinSCP for copying, and its **Commands > Open in PuTTY** option, or ask IT to enable OpenSSH Client.
No such file or directory after unzip or cd	The zip isn't in /root. Repeat Chapter 8, then type **ls /root** to see what's there.
Preflight doesn't say PASS, or the installer says **BPI image marker missing** or **Missing /dev/...**	The card wasn't written with the 3.13 system. Repeat Chapter 5 with a fresh download.
Permission denied running **./install-master.sh**	Type **cd /root**, delete the folder with **rm -r mk-timeclock-lite-RC3-cleanup6**, and unzip again with the unzip command.
Display stays blank after install	Wires 7 to 10 (pins 9, 1, 5, 3) and the I2C change in Chapter 2.
Cards are never read (no beep, nothing on screen)	Wires 1 to 6. 3.3V must be on pin 17. Antenna side facing the front.
No beep	Buzzer red on pin 7, black on pin 6.
Browser warning on the clock's page	Normal until IT installs a certificate (Chapter 10).

> **Write this down for each clock**
> Primary or Secondary · location · IP address · date installed. Keep the ROOT_PASSWORD in a safe place.

Still stuck? Contact iCube Development with a photo of the inside of the case and the last lines of any error message.

TEMPLATE

Wall drilling template

Print at **100 % / Actual size**. Check the 80 mm dimension with a ruler before drilling. Shows the outline of the back of the clock as hung.

